

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

5th Semester of B. Pharm. Examination

University Theory Examination Nov. 2016

PH304 Pharmaceutical Biochemistry- I

Date: 25.11.2016 , Friday

Time: 10.00 a.m to 1.00 p.m

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q 1 Attempt any TWO of the following

- A Draw a labeled diagram of prokaryotic and eukaryotic cells and provide a difference between them. 10
- B 1. Define and classify enzymes. Give significance of each class of enzymes and provide at least one example with its role. 7
2. Differentiate competitive and non-competitive enzyme inhibition. 3
- C Explain various factors regulating enzymatic reactions. 10

Q 2 Attempt any TWO of the following

- A Add a detailed note on β - Oxidation of fatty acids. 10
- B Draw a pathway for biosynthesis of cholesterol and explain its regulation. 10
- Explain following terminologies in brief:
- C 1. Autotrophic organisms
2. Ketogenesis
3. Prosthetic group
4. Coenzyme
5. Triacylglycerols 10

SECTION- II

Q 3 Attempt any TWO of the following

- A 1. Describe the process of transport processes across cell membrane. 6
2. Classify proteins with examples. 4
- B 1. Write a note on pentosephos phate pathway with chemical reactions. 6
2. Write a note on Monosaccharides. 4

C Explain following terminologies in brief:

1. Gluconeogenesis
2. Glycogenolysis
3. Non-essential amino acids
4. Derived proteins
5. Heteropolysaccharide

10

Q 4 Attempt any TWO of the following

- A
1. Describe Insulin and its importance in carbohydrate metabolism.
 2. Enumerate different structures of proteins. Write about β - sheet.

6

4

- B
1. Write a note on functional and chemical classification of amino acids.
 2. Draw overview of TCA cycle and give energetics calculation in it.

6

4

- C
1. Write a note on various pathways of gluconeogenesis.
 2. Write a note on function of Proteins.

6

4